

## Highlights

- Risk-aware BO uses learned feasibility to explore phase-stability boundaries
- 48 HEAs from Al-V-Cr-Mn-Fe-Co-Ni-Cu synthesized over three closed-loop iterations
- V drives both yield strength ( $r=0.84$ ) and sigma-phase failure ( $r=0.54$ )
- V-rich FCC alloys reach 1480 MPa UTS at 50% elongation
- Optima cluster on KKT-active boundary. Hard filtering would exclude them